Augmented Reality Interface for Direct Robotic Welding Path Programming

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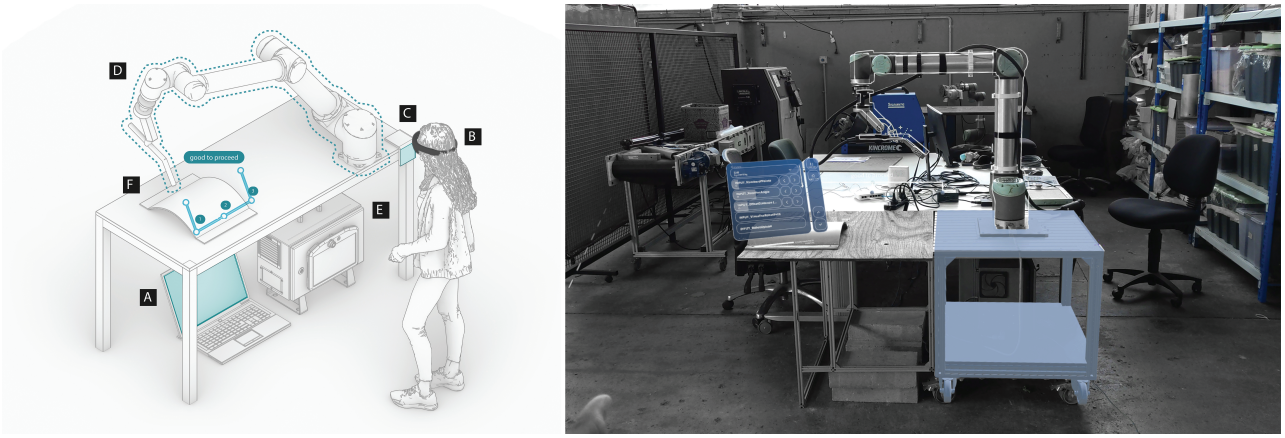


Figure 1: [Left] Proposed AR-assisted robotic welding setup. (A) Workstation, (B) Microsoft HoloLens 2, (C) Visual marker, (D) UR10, collaborative robot, and (E) Control box. [Right] Successful tack welding path preview within the proposed AR interface, illustrating trajectory validation.

Abstract

Programming robotic welding tasks remains challenging for non-experts due to the complexity of traditional teach-pendant and CAD-based methods. Augmented Reality (AR) offers a promising alternative by enabling intuitive, spatially contextualised interaction. Prior work has focused primarily on seam welding through Programming-by-Demonstration (PtD), with limited attention to tack welding tasks. We present a proof-of-concept AR interface that supports interactive definition of both seam and tack welding paths using gesture-based spatial input on a physical workpiece. Our system provides real-time reachability feedback and supports a human-in-the-loop (HITL) planning workflow. An informal evaluation with research team members suggests feasibility, though challenges such as spatial anchoring drift remain. We contribute: (1) an AR-based prototype workflow for tack and seam welding path planning, and (2) an interactive interface leveraging real-time spatial feedback and natural gestural input.

Index Terms: Augmented Reality (AR), Head-Mounted Display (HMD), Human-Robot Interaction (HRI), Robotic Welding, System Design.

1 Introduction and Related Work

Welding is an essential, yet physically demanding and hazardous profession, requiring significant precision and expertise [3]. Despite its critical role in manufacturing and construction settings, industries face a steady decline in skilled welders, driven largely by occupational health risks and physical strain [1]. In recent years, robotic welding systems have been proposed as viable solutions to improve worker safety, productivity and consistency. However, their adoption remains limited due to the complexity and specialised robotics expertise required, which most skilled welders lack. Furthermore, conventional robotic programming for welding tasks often involves using teach pendant or CAD-based programming, which are time consuming, and require accurate representation of the virtual environment [6]. Augmented reality (AR) has recently emerged as a promising solution to overcome these limitations, as it simplifies robotic welding programming through intuitive, spatially contextualised interfaces [5]. AR has been deployed alongside Programming-through-Demonstration (PtD) techniques for robotic seam welding. As demonstrated in [5], the authors use this approach to support non-expert users to define welding paths. It is important to note that the accuracy of PtD heavily depends on the quality of the demonstration and the capability of the motion capture tool. Recognising this limitation, [4] have explored this interaction with path optimisation strategies to enhance the precision of the demonstrations.

Although existing research on AR-supported welding is primarily associated with PtD , recent advances in computational programming allow us to diverge from this paradigm. By leveraging these computational capabilities, we propose an AR-based method for direct definition and manipulation of robotic welding paths. In parallel, we observe that the literature predominantly focuses on seam welding, while tack welding remains relatively unexplored despite its significant role in the assembly process. Our work addresses this gap by applying our computationally driven AR approach specifically to tack welding.

In this work, we present a proof-of-concept AR-based system that allows users to intuitively define both tack and seam welding paths without robotics expertise. To the best of our knowledge, this study is the first to specifically address AR-assisted tack welding, thereby expanding the scope of AR-supported human-robot interaction (HRI) research into a crucial yet underexplored area.

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2 System Design

2.1 System Architecture: Hardware

The hardware system comprises three main components: an AR-HMD, a cobot, and a Windows PC (illustrated in Figure 1). Microsoft HoloLens 2 was used to facilitate immersive visualisation and hands-free, spatial interaction with weld paths through the Fologram [2] application. The robotic workcell includes a collaborative robot (cobot), UR10 CB-series, which executes the welding path defined by the users.

The workstation and cobot are connected via an Ethernet cable to ensure reliable transmission of robot commands. At the same time, the AR-HMD communicates with the workstation over a stable local Wi-Fi network. The AR visual environment is spatially anchored on the cobot's physical workspace via a visual marker.

2.2 System Architecture: Software

The system is built on *Grasshopper*, a visual programming platform embedded in Rhino 3D. Weld points and seam paths are generated using geometry-based definitions, and robot trajectories are planned through the Robots plugin, which supports the simulation and export of URScript-compatible code.

Fologram [2] bridges the AR and programming environments by streaming the Grasshopper model to the *HoloLens 2* in real-time, allowing users to view and interact with weld plans overlaid on the physical workpiece. All parameter changes in the AR environment are synchronised with the underlying Grasshopper model.

2.3 Interaction Workflow

Users interact with the system through an AR interface rendered on the *Microsoft HoloLens 2*. An on-screen menu allows users to switch between tack and seam welding modes. Within each mode, users can adjust design parameters such as (A) welding speed, (B) welding angle, (C) offset from the metal workpieces, and (D) resolution (number of target points). In addition, users can define the number of tack welds while in the tack welding mode.

Weld paths are defined by manipulating the virtual widgets using pinch-and-drag gestures. These widgets are directly associated with robot poses and tool orientations. As users adjust cobot waypoint(s), a real-time trajectory preview is rendered in the visual environment, providing real-time visual feedback on the planned toolpath.

The AR environment also incorporates colour-coded visual cues and text annotation (refer to Figure 2) to indicate reachability constraints, helping users detect and correct infeasible paths before execution. Once validated, users transmit the command directly from the AR interface to the cobot, completing the planning execution loop.

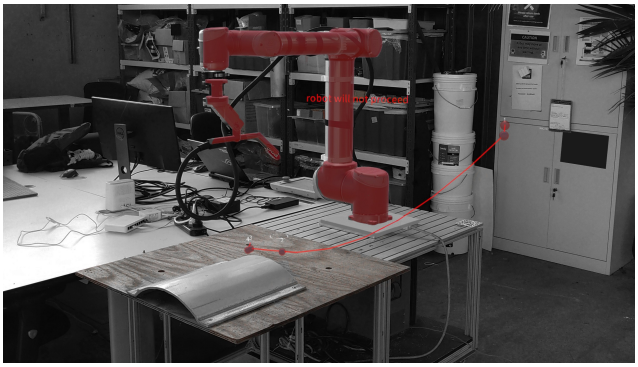


Figure 2: Screenshot of AR interface preview highlighting unreachable weld paths and robot meshes in red with text annotations indicating reachability constraints

3 Results and Discussion

An informal evaluation of the proof-of-concept was conducted with three participants (N=3) including an author and two colleagues. Each participant completed the tack and seam welding task setup in approximately 5-10 minutes without prior training. Spatial anchoring remained relatively stable, but drifted noticeably during longer trails or when the visual marker was not within the field of view (FOV) of the HoloLens. The participants found that AR-based gesture interactions were intuitive, although occasional precision errors occurred when adjusting robot waypoints.

These early findings suggest the system's potential for streamlining robotic welding setup, particularly in environments where rapid prototyping is critical. Although geometry-based interactions effectively supported tack welding, seam welding may benefit more from traditional AR-supported PtD as shown in [4, 5]. Given that welding practice involves significant tacit knowledge, PtD could capture nuances that our proposed approach may overlook. Nonetheless, features such as real-time reachability feedback remain valuable for users with limited robotics expertise.

4 Conclusion

This poster presented a proof-of-concept system that integrates an AR-based interaction for tack and seam welding tasks. By allowing users to define weld paths directly within a spatially anchored AR environment using gestural inputs, the system offers a more simplified and flexible approach to robot programming.

Future work could adopt a more user-centered design approach by involving skilled welders in iterative system development to ensure that the developed system meets their practical needs. We also plan to conduct formal studies to evaluate task efficiency, learning curves, and cognitive load.

Supplemental Materials

Supplementary materials are available on Github at https://github.com/LoyWeiWin/Pub_ISMAR_RoboticWelding, released under a GPL V3 license. In particular, they include (1) figure images in full resolution, (2) low resolution of the videos, (3) a full version of this article, and (4) the code developed.

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References

[1] M. Couto, M. R. Petry, and M. F. Silva. A Study of Virtual Reality Applied to Welder Training. In M. E. Auer, U. R. Cukierman, E. Vendrell Vidal, and E. Tovar Caro, eds., *Towards a Hybrid, Flexible and Socially Engaged Higher Education*, pp. 116–127. Springer Nature Switzerland, Cham, 2024. doi: 10.1007/978-3-031-52667-1_13 1

[2] J. Gwyllim, N. Cam, and v. d. B. Nick. Fologram for grasshopper. https://fologram.com/, 2019. 2

[3] Y.-S. Kim, C.-H. Lee, and Y.-K. Kim. Effect of heat input on microstructure and mechanical properties of austenitic stainless steel welds. *Materials Science and Engineering: A*, 396:1–9, 2005. doi: 10.1016/j.msea.2004.11.063 1

[4] G. Liu, W. Sun, and P. Li. Motion capture and AR based programming by demonstration for industrial robots using handheld teaching device. *Scientific Reports*, 14(1):23259, 2024. doi: 10.1038/s41598-024-73747-4 1, 2

[5] S. K. Ong, A. Y. C. Nee, A. W. W. Yew, and N. K. Thanigaivel. AR-assisted robot welding programming. *Advances in Manufacturing*, 8(1):40–48, 2020. doi: 10.1007/s40436-019-00283-0 1, 2

[6] Z. Zhang, B. Wang, X. Zhao, J. Li, and G. Xiong. A hybrid offline programming approach for robotic welding of multi-pipe intersections. *Robotics and Computer-Integrated Manufacturing*, 83:102599, 2023. doi: 10.1016/j.rcim.2023.102599 1